1.5 Monitorización del tráfico de red con TCPDUMP

Tcpdump es una herramienta de línea de comandos para capturar y analizar el tráfico de red. Es una herramienta que puede ser utilizada para una variedad de propósitos, incluyendo:

- **Diagnóstico de problemas de red**: *Tcpdump* puede ser utilizado para identificar problemas de red, como pérdida de paquetes, congestionamiento o ataques de red.
- **Investigación de seguridad**: *Tcpdump* puede ser utilizado para investigar incidentes de seguridad, como ataques de malware o violaciones de datos.
- **Análisis de rendimiento**: *Tcpdump* puede ser utilizado para analizar el rendimiento de la red, como la velocidad de transferencia de datos o el uso de ancho de banda.

Ejemplos de uso de tcpdump:

Para mostrar el listado de interfaces de red disponibles:

```
usuario@ubuntu-20:~$ tcpdump -D
1.enp0s3 [Up, Running]
2.lo [Up, Running, Loopback]
3.any (Pseudo-device that captures on all interfaces) [Up, Running]
4.bluetooth-monitor (Bluetooth Linux Monitor) [none]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
usuario@ubuntu-20:~$
```

Para capturar todo el tráfico de una interfaz:

```
usuario@ubuntu-20:~$ sudo tcpdump -i enp0s3
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
10:51:56.939719 IP ubuntu-20.ntp > prod-ntp-4.ntp1.ps5.canonical.com.ntp: NTPv4, Client, length
10:51:56.941473 IP ubuntu-20.50808 > 172.23.136.5.domain: 63329+ [1au] PTR? 57.190.125.185.in-a
10:51:57.030970 IP prod-ntp-4.ntp1.ps5.canonical.com.ntp > ubuntu-20.ntp: NTPv4, Server, length
10:51:57.554511 IP 172.23.136.5.domain > ubuntu-20.50808: 63329 2/3/1 PTR prod-ntp-4.ntp1.ps5.c
7)
10:51:57.555743 IP ubuntu-20.37984 > 172.23.136.5.domain: 27538+ [1au] PTR? 15.2.0.10.in-addr.a
10:51:57.633099 IP 172.23.136.5.domain > ubuntu-20.37984: 27538 NXDomain 0/1/1 (128)
10:51:57.633494 IP ubuntu-20.37984 > 172.23.136.5.domain: 27538+ PTR? 15.2.0.10.in-addr.arpa. (
10:51:57.640020 IP 172.23.136.5.domain > ubuntu-20.37984: 27538 NXDomain 0/1/0 (117)
10:51:57.641587 IP ubuntu-20.32990 > 172.23.136.5.domain: 17609+ [1au] PTR? 5.136.23.172.in-add
10:51:57.757766 IP 172.23.136.5.domain > ubuntu-20.32990: 17609 NXDomain 0/1/1 (131)
10:51:57.758065 IP ubuntu-20.32990 > 172.23.136.5.domain: 17609+ PTR? 5.136.23.172.in-addr.arpa
10:51:57.763992 IP 172.23.136.5.domain > ubuntu-20.32990: 17609 NXDomain 0/1/0 (120)
10:51:59.935493 IP ubuntu-20.ntp > prod-ntp-5.ntp1.ps5.canonical.com.ntp: NTPv4, Client, length
```

Para capturar tráfico de un protocolo específico:

```
usuario@ubuntu-20:~$ sudo tcpdump -i enp0s3  tcp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
10:53:31.275791 IP ubuntu-20.59162 > 82.221.107.34.bc.googleusercontent.com.http: Flags [S], seq 3593790407, win 64240, options [mss
1790822752 ecr 0,nop,wscale 7], length 0
10:53:31.283835 IP 82.221.107.34.bc.googleusercontent.com.http > ubuntu-20.59162: Flags [S.], seq 8064001, ack 3593790408, win 65535,
, length 0
10:53:31.283865 IP ubuntu-20.59162 > 82.221.107.34.bc.googleusercontent.com.http: Flags [.], ack 1, win 64240, length 0
10:53:31.284434 IP ubuntu-20.59162 > 82.221.107.34.bc.googleusercontent.com.http: Flags [P.], seq 1:323, ack 1, win 64240, length 322
cal.html HTTP/1.1
10:53:31.284807 IP 82.221.107.34.bc.googleusercontent.com.http > ubuntu-20.59162: Flags [.], ack 323, win 65535, length 0
10:53:31.295850 IP 82.221.107.34.bc.googleusercontent.com.http > ubuntu-20.59162: Flags [P.], seq 1:299, ack 323, win 65535, length 2
200 OK
10:53:31.295877 IP ubuntu-20.59162 > 82.221.107.34.bc.googleusercontent.com.http: Flags [.], ack 299, win 63942, length 0
10:53:31.490169 IP ubuntu-20.59168 > 82.221.107.34.bc.googleusercontent.com.http: Flags [S], seq 3328854718, win 64240, options [mss
1790822966 ecr 0,nop,wscale 7], length 0
10:53:31.498263 IP 82.221.107.34.bc.googleusercontent.com.http > ubuntu-20.59168: Flags [S.], seq 8128001, ack 3328854719, win 65535,
, length 0
10:53:31.498528 IP ubuntu-20.59168 > 82.221.107.34.bc.googleusercontent.com.http: Flags [.], ack 1, win 64240, length 0
10:53:31.498997 IP ubuntu-20.59168 > 82.221.107.34.bc.googleusercontent.com.http: Flags [P.], seq 1:325, ack 1, win 64240, length 324
s.txt?ipv4 HTTP/1.1
```

Para capturar tráfico de un interfaz, a un host destino y el puerto HTTPS sería de la siguiente forma, incluyendo el operador lógico *and*.

```
usuario@ubuntu-20:~$ sudo tcpdump -i enp0s3 dst 104.16.57.101 and port 443
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
21:45:54.566242 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [P.], seq 769729333:769729413, ack 7172633, win 6
21:45:54.567114 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [P.], seq 80:546, ack 1, win 62780, length 466
21:45:54.576190 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [.], ack 83, win 62780, length 0
21:45:59.675267 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [P.], seq 546:626, ack 83, win 62780, length 80
21:45:59.675401 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [P.], seq 626:2227, ack 83, win 62780, length 160
21:45:59.676149 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [P.], seq 2227:4648, ack 83, win 62780, length 24
21:45:59.681924 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [.], ack 118, win 62780, length 0
21:45:59.686616 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [.], ack 200, win 62780, length 0
21:46:00.028053 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [P.], seq 4648:4728, ack 200, win 62780, length 8
21:46:00.028943 IP ubuntu-20.50094 > 104.16.57.101.https: Flags [P.], seq 4728:5193, ack 200, win 62780, length 4
```

Para capturar tráfico de un puerto específico:

```
usuario@ubuntu-20:~$ sudo tcpdump -i enp0s3  tcp port 443
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
10:55:15.371311 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [S], seq 1759487262, win 64240, options [mss 1460,sackOK,TS val 212409
ale 7], length 0
10:55:15.379905 IP 104.26.11.240.https > ubuntu-20.43530: Flags [S.], seq 23424001, ack 1759487263, win 65535, options [mss 1460], leng
10:55:15.379928 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [.], ack 1, win 64240, length 0
10:55:15.383202 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [P.], seq 1:1161, ack 1, win 64240, length 1160
10:55:15.383842 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [P.], seq 1161:1167, ack 1, win 64240, length 6
10:55:15.383910 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [P.], seq 1167:1337, ack 1, win 64240, length 170
10:55:15.383932 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], ack 1161, win 65535, length 0
10:55:15.384804 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], ack 1167, win 65535, length 0
10:55:15.384804 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], ack 1337, win 65535, length 0
10:55:15.396549 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], seq 1:2921, ack 1337, win 65535, length 2920
10:55:15.396586 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [.], ack 2921, win 62780, length 0
10:55:15.397669 IP 104.26.11.240.https > ubuntu-20.43530: Flags [P.], seq 2921:2989, ack 1337, win 65535, length 68
10:55:15.397680 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [.], ack 2989, win 62780, length 0
10:55:15.433355 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [P.], seq 1337:1395, ack 2989, win 62780, length 58
10:55:15.433867 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [P.], seq 1395:1565, ack 2989, win 62780, length 170
10:55:15.434133 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [P.], seq 1565:1604, ack 2989, win 62780, length 39
10:55:15.434454 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], ack 1395, win 65535, length 0
10:55:15.434454 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], ack 1565, win 65535, length 0
10:55:15.434454 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], ack 1604, win 65535, length 0
10:55:15.434934 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [P.], seq 1604:1628, ack 2989, win 62780, length 24
10:55:15.435174 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], ack 1628, win 65535, length 0
10:55:15.435256 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [F.], seq 1628, ack 2989, win 62780, length 0
10:55:15.436306 IP 104.26.11.240.https > ubuntu-20.43530: Flags [.], ack 1629, win 65535, length 0
10:55:15.441545 IP 104.26.11.240.https > ubuntu-20.43530: Flags [P.], seq 2989:3517, ack 1629, win 65535, length 528
10:55:15.441570 IP ubuntu-20.43530 > 104.26.11.240.https: Flags [R], seq 1759488891, win 0, length 0
```

Para analizar el tráfico de red DNS

```
usuario@ubuntu-20:~$ sudo tcpdump -i enp0s3  udp port 53
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
10:56:21.371902 IP ubuntu-20.46389 > 172.23.136.5.domain: 33957+ [1au] AAAA? youtube-ui.l.google.com. (52)
10:56:21.373226 IP ubuntu-20.55886 > 172.23.136.5.domain: 51428+ [1au] PTR? 5.136.23.172.in-addr.arpa. (54)
10:56:21.390301 IP 172.23.136.5.domain > ubuntu-20.46389: 33957 4/0/1 AAAA 2a00:1450:4003:80c::200e, AAAA 2a00:1450:4003
811::200e, AAAA 2a00:1450:4003:806::200e (164)
10:56:21.462368 IP 172.23.136.5.domain > ubuntu-20.55886: 51428 NXDomain 0/1/1 (131)
10:56:21.462562 IP ubuntu-20.55886 > 172.23.136.5.domain: 51428+ PTR? 5.136.23.172.in-addr.arpa. (43)
10:56:21.469164 IP 172.23.136.5.domain > ubuntu-20.55886: 51428 NXDomain 0/1/0 (120)
10:56:21.470113 IP ubuntu-20.37436 > 172.23.136.5.domain: 17201+ [1au] PTR? 15.2.0.10.in-addr.arpa. (51)
10:56:21.584570 IP 172.23.136.5.domain > ubuntu-20.37436: 17201 NXDomain 0/1/1 (128)
10:56:21.584934 IP ubuntu-20.37436 > 172.23.136.5.domain: 17201+ PTR? 15.2.0.10.in-addr.arpa. (40)
10:56:21.590170 IP 172.23.136.5.domain > ubuntu-20.37436: 17201 NXDomain 0/1/0 (117)
10:56:31.282132 IP ubuntu-20.37086 > 172.23.136.5.domain: 39556+ [1au] A? cloudflareinsights.com. (51)
10:56:31.289167 IP 172.23.136.5.domain > ubuntu-20.37086: 39556 2/0/1 A 104.16.56.101, A 104.16.57.101 (83)
10:56:31.500285 IP ubuntu-20.43151 > 172.23.136.5.domain: 24710+ [1au] A? fonts.gstatic.com. (46)
10:56:31.508138 IP 172.23.136.5.domain > ubuntu-20.43151: 24710 1/0/1 A 142.250.200.67 (62)
10:56:34.550574 IP ubuntu-20.45274 > 172.23.136.5.domain: 13402+ [1au] A? googleads.g.doubleclick.net. (56)
10:56:34.557515 IP 172.23.136.5.domain > ubuntu-20.45274: 13402 1/0/1 A 142.250.200.130 (72)
```

Ejercicio 8. Captura usando Tcpdump y volcando la información al fichero la operación de hacer ping al ordenador de un compañero de clase.

Es decir:

1. Abrir un terminal y hacer ping al host de un compañero → Pantallazo
2. Abrir un terminal y capturar con tcpdump el tráfico de ping y para esa IP y generar la salida a un fichero de salida (**tunombre.pcap**) que será consultado posteriormente. → Pantallazo del comando Tcpdump

Solución

```
usuario@ubuntu-20:~$ sudo tcpdump -i enp0s3  dst 192.168.1.95 and  icmp  -w juan.pcap
```

3. Abrir con wireshark el archivo pcap generado → Pantallazo como el que se acompaña

